How the boat model works

A plain-English tour of `explore.ipynb`

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The question

We want a boat that covers two miles as fast as possible. This document explains the model in `explore.ipynb` that tries to find one.

The trouble is that everything depends on everything else. A faster boat needs more power. More power needs a bigger motor and a bigger battery. Those make the boat heavier. A heavier boat needs more power to go the same speed. So you can't choose the motor first and the battery second. You have to choose them together, along with the shape of the hull, and then see whether the whole thing works.

That is what the model does. You give it a complete design. It works out how much the boat weighs, where the weight sits, how hard the water pushes back, how much power that takes and how much battery that uses. Then it checks the design against twenty-four rules. If the design passes, the model tells you how long the race would take.

One warning. Nothing here has been checked against a real boat. The motor is real data from the supplier's spec sheet, but the battery cells, the speed controller's efficiency and the propeller's blade losses and weight are placeholders. So read the answers as "what the model believes," not "what the boat will do." The model is more useful for something else: it shows which limits bite first, and which choices matter.

What we get to choose

There are eleven choices, of five kinds. The shape of the hull: length, beam (the width), deadrise (the angle of the V on the bottom) and depth. Where the payload and the battery sit along the hull. The motor: which one winding of which SSS brushless motor from the supplier's spec sheet. The propeller: its pitch and its diameter. And the battery: how many cells go in series, and how many of those strings go in parallel. Series sets the voltage. Parallel sets the capacity, and how much current the pack can supply.

Eight of the eleven vary continuously between bounds. The two battery counts and the motor are whole numbers. The motor is a position in a list of 230 windings, ordered from the smallest motor to the largest. Each winding brings its own real numbers from the sheet: how fast it turns per volt, its current and voltage limits, its resistance, its power rating and its weight.

Table 1: Searched decision variables. Bounds are the notebook's `BOUNDS`; the baseline is the default `Design`.

Symbol	Code name	Unit	Type	Bounds	Baseline	Group
L	`length_ft`	ft	continuous	[3, 10]	8.5	hull
b	`beam_in`	in	continuous	[10, 40]	16	hull (chine beam)
β	`deadrise_deg`	deg	continuous	[6, 16]	10	hull
D	`depth_in`	in	continuous	[8, 20]	12	hull
ξ_p	`payload_x`	–	continuous	[0.25, 0.50]	0.30	placement (fraction of L forward of transom)
ξ_b	`battery_x`	–	continuous	[0.25, 0.45]	0.35	placement (fraction of L forward of transom)
μ	`motor`	–	integer	$\{0, \ldots, 229\}$	222	SSS winding: position in the catalog (baseline: 56154-V2 5Y)
p	`pitch_in`	in	continuous	[4, 16]	10	propeller pitch
d_p	`prop_diam_in`	in	continuous	[3, 10]	6	propeller diameter
S	`series`	–	integer	$\{12, 13, 14\}$	14	battery series cells
n_p	`parallel`	–	integer	$\{1, \ldots, 12\}$	6	battery parallel strings

Speed is missing from the list, and you might expect it to be there. It isn't, because for any given design there is a fastest speed at which every rule still holds, and nobody would run slower. So we don't ask the search to pick a speed. For each design, the model finds that speed itself and uses it. This turns the question into a simpler one: which design has the highest legal top speed?

The notebook has one more field that the search leaves alone: thrust angle. It does affect the balance, but it is held at zero. Pitch and diameter used to be left out too, because propeller efficiency was a fixed constant and they could not change the answer. The propeller now has a small model (next section and Appendix A), so they matter and the search chooses them.

How the model works out the rest

Think of it as a chain. Each link uses what came before. Figure 1 shows the order.

Weight and balance. Start with weight. The hull's weight is the area of its mesh (bottom, sides, deck and transom, including the tapered bow) times the weight of the fiberglass per square foot, plus a fixed allowance for structure. The battery's weight is the number of cells times the weight of one cell, plus 15% for the case and wiring. The motor's weight is the weight the supplier lists for that motor, so a bigger, stronger motor costs boat weight. That stops the search from simply picking the biggest motor. The propeller's weight grows with the cube of its diameter, so a bigger propeller is heavier, and it sits at the very back of the hull, which pulls the center of gravity aft. Add the 30 lb payload and 8.8 lb of fixed equipment, and you have the total.

Each part sits somewhere along the hull and at some height above the keel. Averaging by weight gives the center of gravity. Where it sits front to back matters a lot, as you'll see.

Floating. Next, how deep the boat sits when it's stopped. The notebook builds a watertight triangle mesh of the hull – the V-bottom prism, with the chine beam tapering linearly to the stem

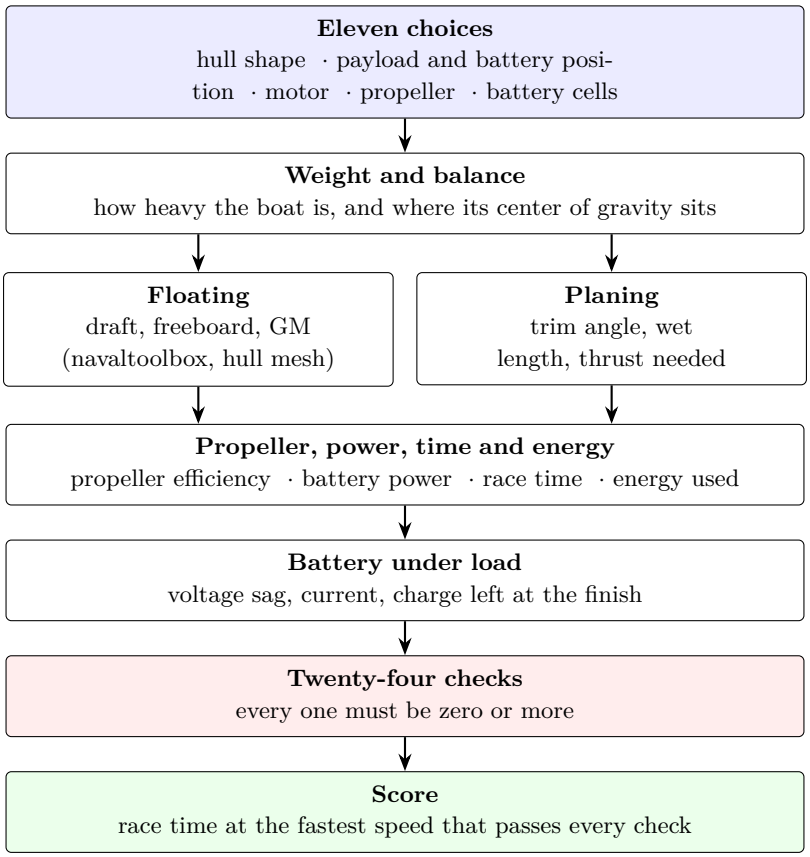


Figure 1: What happens to one design at one speed.

over the forward quarter of the length – and a naval-architecture library, navaltoolbox, finds the draft at which the displaced water weighs as much as the boat. Subtract the draft from the hull's depth and you get freeboard: how much hull is above the water. A planing boat rides higher once it's moving, so this is the cautious number.

The same mesh gives the metacentric height GM, the standard measure of how stiffly a boat resists rolling: push it over a little, and GM says how hard it rights itself. The model requires a minimum GM at rest, so a hull too narrow to stand up straight is now infeasible rather than merely unwise. For the checks the boat is held level; its at-rest pitch equilibrium (it trims slightly by the stern, because the center of gravity sits aft of the center of buoyancy) is computed for the charts but not constrained.

Planing. Now the hard part. At speed, a boat like this lifts up and skims across the water. Where it settles is a balance. The water has to push up with a force equal to the boat's weight, and it has to do that without tipping the bow up or down. The angle the boat settles at is called trim. How much of the hull stays wet depends on trim, and the drag depends on how much is wet.

We don't work this out ourselves. A library called OpenPlaning does it, using a method published by Savitsky. We give it the weight, the center of gravity, the beam, the deadrise and the speed. It gives back the trim, how much of the hull is wet, and the thrust the propeller must supply to keep going. For the default boat at about 21 mph the thrust is 102 newtons. At this size most of the drag is just water rubbing along the bottom.

The propeller. Thrust times speed is the useful work the boat does each second. The propeller wastes some of it, and how much depends on the propeller. We start from the textbook ideal: a propeller is a disk that accelerates water backwards, and the more heavily that disk is loaded (more thrust through less area, at a given speed), the more energy goes into the wash instead of the boat. A larger diameter spreads the same thrust over more water and wastes less. Two placeholder factors sit on top of the ideal: blade losses, and a penalty when the pitch is far from about 1.2 times the diameter. For the default boat the result is about 66%, close to the 65% constant it replaced. The propeller also has to turn at a speed set by the boat's speed and its pitch (a coarser pitch turns slower), and the tip of the blade has a speed limit, because a tip moving too fast makes bubbles that ruin the thrust. That limit is what stops the search from simply choosing a huge, fast propeller. The propeller must also be no deeper than the hull.

Power, time and energy. The motor wastes some more, and how much comes from the motor's own numbers. The propeller has to turn at the rpm that the boat's speed and pitch demand, and it asks for a torque (power divided by angular speed). A motor makes torque from current, and the newton-metres per amp are fixed by how fast it turns per volt. That current also runs through the winding's resistance and heats it, and a little current is always spent turning the motor with no load. So the motor's efficiency is worked out, not assumed. For the default boat, whose propeller turns slowly and so asks for a lot of torque, the motor is only about 63% efficient. A speed controller between the battery and the motor loses another 3%. Add 30 watts for electronics. What comes out is the power the battery must supply. The default boat needs about 2.4 kilowatts.

Race time is distance divided by speed. We use speed over the ground, which is the speed through the water minus a 1 mph current, and add 20 seconds of allowance. Energy used is power times that time. The 20 seconds is charged at full cruise power, which is simple rather than realistic.

The battery. The battery has two quirks. First, the competition caps pack voltage at 55.5 volts when fully charged. Fourteen cells in series would reach 58.8 volts when full, so the model charges the pack only as far as the cap allows. For the default boat that is 75.5%. This is computed, not chosen.

Second, a battery's voltage sags when you draw current from it, because of its internal resistance. More strings in parallel share the load and sag less. The model solves for the voltage under load, then the current, then how much charge and what cell voltage remain at the finish.

What the boat has to obey

The model checks twenty-four things. Each is a number that has to be zero or more. We call it a margin: how much room is left before the rule is broken. A design is feasible if all twenty-four margins are nonnegative. They come in four kinds. The exact formulas are in Appendix C; here is what they mean.

Competition rules (three). Freeboard at rest must be at least 3 inches, plus half the 4-inch wave height, so 5 inches in all. The full-charge pack voltage must be at most 55.5 volts. And the boat must make headway against the current.

Being a boat (two). The hull must be deeper than the height of its chine, the edge where the bottom meets the sides. Otherwise there is no hull above the chine to speak of. And the propeller must be no deeper than the hull.

Hardware and balance (twelve). A fully charged pack must not exceed the motor winding's voltage rating on the sheet. The race may use at most 70% of the energy in the pack at the start. Shaft power must stay under 80% of the motor's continuous power rating, the battery's current under 80% of the battery's rating, and the motor's current under 80% of the winding's maximum current. The loaded battery must also be able to supply the voltage the motor needs at that speed: its back-voltage plus the drop across its own resistance. That 20% cushion is a design margin. The pack must be able to deliver the power at all; past a certain draw the voltage equation has no solution. Each cell must stay above 3.0 volts at the finish. The propeller's blade tip must stay under 150 feet per second. The center of gravity must sit between 30 and 45 percent of the hull's length from the back. And the metacentric height at rest must be at least 0.05 m – a placeholder stability floor, but enough to stop the search from taking an absurdly narrow hull for free.

Staying inside the method (seven). Savitsky's formulas were fitted to experiments over a limited range, and outside it the library is guessing. So the model treats the edge of that range as a wall. The solver must find a balance at all. Trim must be between 2 and 15 degrees. The wetted length can be at most four beam-widths. A speed number built from speed and beam must lie between 0.6 and 13. And the wetted keel can't be longer than the hull.

Eight of the twenty-four don't depend on speed at all: freeboard, stability, hull depth, propeller fit, both voltage checks and both center-of-gravity checks. They decide whether a design has any legal speed. The other sixteen get harder as speed rises, and together they set the top speed.

What we minimise

Race time, and nothing else. There is no cost term, and no goal for weight beyond the limits above.

For one design, the model finds the top speed like this. Try 8 mph, then 12, 16 and so on up to 60. Remember the highest speed at which all twenty-four checks pass. Then narrow in on the edge by repeated halving until it is pinned down to a tenth of a mile per hour. The race time at that speed is the design's score. (Inside the search the scan is coarser, to be quicker.)

If no speed passes, the design gets a score of 1000 plus a measure of how badly it breaks the rules at 20 mph. That nudges the search toward designs that work.

To search over designs we use differential evolution: keep a population of candidate designs, mix and mutate the good ones, keep the improvements. The notebook runs it with three different random seeds. It is a good way to explore, but it doesn't prove that no better design exists. If three seeds agree, that is reassuring, not conclusive.

Part II of the notebook also tries other ways of searching the same space (random search, dual annealing and a local search), and two other objectives. Both of those fix the speed at a target of 25 mph and ask a different question of the same twenty-four checks. One asks for the lightest boat that can run that speed. The other asks for the boat that uses the least energy over the course at that speed. An infeasible design scores 1000 plus 100 times how badly it breaks the rules. Race time stays the main objective, and the rest of this document is about it.

An example

Take the default design: 8.5 ft long, 16-inch beam, 10-degree deadrise, an SSS 56154-V2 5Y motor (400 Kv, 14 kW continuous, 3.9 lb), a 6-inch propeller of 10-inch pitch, 14 cells in series and 6 strings. It weighs 78 lb. Its top speed is 21.25 mph, and its race time is 376 seconds, about six minutes. The project's benchmark is seven.

Table 2: Calculated values for the baseline design at $v^\star = 21.25$ mph.

Quantity	Value	Unit	Quantity	Value	Unit
hull mass m_{hull}	19.44	lb	thrust T	101.9	N
battery mass m_{bat}	14.88	lb	running trim τ	2.003	deg
motor mass m_{mot}	3.92	lb	wetted length / beam λ	3.360	–
total mass m	77.54	lb	beam speed coeff. C_v	4.758	–
LCG/L	0.3465	–	shaft power P_{shaft}	1472	W
pack energy E_{pack}	1512	Wh	bus power P_{bus}	2444	W
start charge s_0	75.5	%	race time T_{race}	375.6	s
V_{oc}	55.5	V	energy used E_{used}	255.0	Wh
static draft	2.43	in	loaded voltage V_{load}	53.91	V
freeboard	9.57	in	bus current I	45.3	A
GM at rest	0.208	m	end charge s_{end}	58.7	%
motor torque Q	5.32	N m	end voltage V_{end}	51.88	V
motor current I_m	227	A	motor efficiency η_m	0.629	–
motor voltage needed V_m	10.3	V	propeller efficiency η_p	0.657	–
I_{lim}	90	A	tip speed v_{tip}	69.1	ft/s
propeller mass m_{prop}	0.50	lb	propeller rpm	2640	rpm
Kv needed (rpm/V_{load})	49	rpm/V			

Now ask what stops it going faster. It isn't the motor's power, which is only 11% of its continuous rating. The motor's current is closer: 227 A against the 256 A it is allowed, because the slow propeller asks for a lot of torque. It isn't the battery, which is using about 63% of its allowed current and about 32% of its allowed energy. What stops it is trim. At 21.25 mph the boat's trim is 2.003 degrees, and the method's floor is 2. Go a little faster and the boat flattens below 2 degrees, where the formulas are no longer trusted.

So the model says the top speed is where its own method stops being valid. That is an uncomfortable answer, because it says nothing about the hardware. Appendix C lists all twenty-four margins for this design.

What to doubt

- **The speed limit comes from the method, not the boat.** Trim binds first, so nothing after it in the chain (drag, power, energy) gets a chance to bind. That is why the made-up numbers there look unimportant. They would matter if the trim floor were relaxed.
- **Stability is now a rule, but a thin one.** The mesh gives a real metacentric height and the model requires a minimum of it, so a hull too narrow to stand up is infeasible. But the floor

itself (0.05 m) is a placeholder, the check is at rest in calm water with the boat held level, and there is still no rule about packaging, ride or handling. When a variable still ends up on its bound, the model is telling you it is missing a rule.

- **Most inputs are placeholders.** The cell data, the speed controller's efficiency, the propeller's blade losses, pitch penalty, weight law and tip-speed limit, the hull's bow taper fraction and required GM, the wave height, the current, the heights of components and the laminate weights. Appendix B marks each one.
- **One check is zero by construction.** The charge level is chosen so the pack voltage lands exactly on 55.5 volts, so for 14 cells the voltage margin is zero up to rounding. It passes only because the arithmetic rounds the right way. A check written as "at least zero" on a number that is zero by design is fragile.
- **The motor data is only as good as the sheet.** It was typed from the supplier's spec sheet. 33 of the 263 windings have a resistance entry that looks like a dropped digit, and they are left out: a winding is dropped when its maximum current times its resistance is under 1% or over 30% of its maximum voltage, because the search would otherwise pick a motor that only looks efficient through a typo. The no-load current is the sheet's 11.1 V value, held constant. The drive is direct, with no gearbox, so the motor turns at the propeller's rpm. A real controller would let the motor run above or below the pack voltage; here it only has to be covered.
- **The propeller model is a sketch.** The structure is physical (ideal disk efficiency falls with loading, tip speed is $\pi d \cdot \text{rpm}$, mass grows with volume), but the blade losses, the preferred pitch-to-diameter ratio, the mass law and the tip-speed limit are guesses, and the rpm still comes from a fixed slip factor rather than from the blade's own thrust curve. Real propellers also have cavitation, ventilation and blade-area effects it ignores. Wave height matters only through freeboard.
- **The speed scan trusts the highest passing speed.** It doesn't check that the speeds below it also pass. A design that passes at 12 and 20 mph but not at 16 would be scored at about 20.
- **Two small quirks.** Turning off the validity checks (`ENFORCE_VALIDITY = False`) lets a design whose planing solve failed count as feasible; the default is on. And the score of 1000 for infeasible designs is only safely worse than feasible ones above about 8.35 mph, since slower boats take more than 1000 seconds.

What to replace next

Real cell data first: resistance, continuous current and the voltage curve. Then confirm the sheet's doubtful resistance entries with the supplier, add a gearbox ratio, the controller's real efficiency, a propeller map in place of the placeholder propeller model, and real hull stations from the CAD hull in place of the linear bow taper. Then the wave height and current for the actual venue, and real component heights. The notebook leaves out transom beam, bow deadrise and keel curve because OpenPlaning can't use them.

A The formulas, in order

These are the exact formulas behind the chain described in the main text, in the order the notebook computes them. Nothing here is chosen. Every value follows from the eleven decisions, the constants in Appendix B and the values before it.

A.1 Hull geometry and mass

The hull is a V-bottom section of chine beam b and deadrise β with vertical sides up to the depth D, prismatic from the transom forward. Over the forward fraction f_b of the length (the bow), the chine half-beam tapers linearly to zero at the stem; the keel and the deck stay level. With $b_{\text{ft}} = b/12$ and $t = \tan\beta$, the chine sits at height

$$h_c = \tfrac{1}{2}\, b_{\text{ft}}\, t \quad \text{[ft]} \qquad (\texttt{chine_height_ft}), \tag{1}$$

and because the section is similar everywhere, the chine height falls with the local half-beam toward the stem. `hull_mesh` builds a watertight triangle mesh of exactly this shape (two stations over the prismatic body, eight over the taper, mirrored port and starboard, closed at transom and stem; outward orientation is asserted through the mesh's signed volume). The shell area is the total area of that mesh – bottom, sides, deck, transom and stem together – and the hull mass is

$$m_{\text{hull}} = \rho_{\text{lam}}\, A_{\text{mesh}} + m_{\text{str}} \quad \text{[lb]} \qquad (\texttt{shell_area_ft2},\ \texttt{hull_mass_lb}). \tag{2}$$

A.2 Battery pack

With $n = S\, n_p$ cells,

$$E_{\text{pack}} = n\, q_c\, V_{\text{nom}}\ \text{[Wh]}, \quad m_{\text{bat}} = n\, m_c(1+\kappa_o)\ \text{[lb]}, \quad R_{\text{pack}} = \frac{S}{n_p}\, r_c\ [\Omega], \quad I_{\text{lim}} = n_p\, I_{c,\max}\ \text{[A]}. \tag{3}$$

Series cells add resistance and parallel strings share it, so more parallel strings reduce voltage sag.

Start-of-race charge (derived, not chosen). The pack starts at the highest state of charge at which its open-circuit voltage stays at or below the voltage rule, capped at 100%:

$$s_0 = \begin{cases} 1 & S\, V_{\text{full}} \le V_{\text{lim}}, \\ \text{OCV}^{-1}\!\left(V_{\text{lim}}/S\right) & \text{otherwise}, \end{cases} \qquad V_{\text{oc}} = S \cdot \text{OCV}(s_0). \tag{4}$$

For the baseline, $S = 14$ gives $14 \times 4.2 = 58.8 > 55.5\,\text{V}$, so the pack is charged only to $s_0 = 75.5\%$ and $V_{\text{oc}} = 55.5\,\text{V}$. For $S = 12, 13$ the pack starts full.

A.3 Motor and propeller

The motor is one winding μ of the catalog (`CATALOG[d.motor]`), read from the supplier's spec sheet (`sss_motors.csv`). The sheet gives its constants: K_v (rpm per volt), maximum current $I_{m,\max}$, maximum voltage $V_{m,\max}$, winding resistance R_m, no-load current I_0, continuous power P_{cont} and weight. Its mass is that weight,

$$m_{\text{mot}} = \frac{\text{weight in grams}}{453.592}\ \text{[lb]} \qquad (\texttt{motor_mass_lb}). \tag{5}$$

The drive is direct: the motor turns at the propeller's rpm (8). To put the shaft power P_{shaft} (17) on the shaft at that rpm, the DC motor model gives

$$\omega = \frac{2\pi\,\text{rpm}}{60}, \qquad Q = \frac{P_{\text{shaft}}}{\omega}, \qquad K_t = \frac{60}{2\pi K_v}, \tag{6}$$

$$I_m = \frac{Q}{K_t} + I_0, \qquad V_m = \frac{\text{rpm}}{K_v} + I_m R_m, \qquad P_{\text{el}} = V_m\, I_m, \qquad \eta_m = \frac{P_{\text{shaft}}}{P_{\text{el}}}. \tag{7}$$

Torque comes from the current beyond the no-load current, and the voltage has to cover the back-voltage rpm/K_v plus the drop across the winding. The no-load current I_0, measured by the supplier at 11.1 V, is held constant. This is `motor_point` in the notebook, and η_m is a result, not an input.

Motor limit modes. The caps the margins use come from `motor_caps`, controlled by the `MOTOR_LIMITS` mode switch. With `"sheet"` (the default, assumed throughout this document) they are the sheet's own maxima, $I_{\text{cap}} = I_{m,\max}$ and $V_{\text{cap}} = V_{m,\max}$. With `"theoretical"` they are derived from the DC model instead: $V_{\text{cap}} = \text{rpm}_{\max}/K_v$, the back-EMF at the sheet's rpm ceiling, and $I_{\text{cap}} = P_{\text{cont}}/V_{\text{cap}} + I_0$, the current that carries the continuous power at that voltage – useful for asking how conservative the sheet is. The propeller has pitch p and diameter d_p (both in inches). Its rpm, tip speed and the motor constant it demands are

$$\text{rpm} = \frac{1056\,v}{p\,\sigma_s}, \qquad v_{\text{tip}} = \pi\,\frac{d_p}{12}\,\frac{\text{rpm}}{60}\ \text{[ft/s]}, \qquad K_v = \frac{\text{rpm}}{V_{\text{load}}}. \tag{8}$$

The motor constant K_v here is the one the propeller *demands*; the winding's own K_v is a property of the catalog entry. The tip speed is limited by the margin `tip_speed` (Appendix C). The propeller's mass scales with diameter cubed,

$$m_{\text{prop}} = m_{\text{ref}}\Big(\frac{d_p}{d_{\text{ref}}}\Big)^3\ \text{[lb]} \qquad (\texttt{prop_mass_lb}), \tag{9}$$

and its efficiency is the ideal actuator-disk (Froude) efficiency at the thrust T and speed $U = 0.44704\,v$, times two placeholder factors:

$$C_T = \frac{T}{\frac{1}{2}\rho\,A_p\,U^2}, \qquad A_p = \frac{\pi}{4}(0.0254\,d_p)^2, \tag{10}$$

$$\eta_p = \underbrace{\frac{2}{1+\sqrt{1+C_T}}}_{\text{ideal disk}} \cdot \eta_b \cdot \eta_{pd}, \qquad \eta_{pd} = \max\Big(0.3,\ 1 - k_{pd}\big(\tfrac{p}{d_p} - \text{pd}_{\text{opt}}\big)^2\Big). \tag{11}$$

This is `prop_efficiency` in the notebook. At fixed thrust, a larger d_p lowers C_T and raises the ideal efficiency. The blade factor η_b stands for blade-section, swirl and tip losses. The model has no thrust curve for the blade: the rpm still comes from the fixed slip factor σ_s, and pitch affects the efficiency only through the ratio p/d_p.

A.4 Mass and balance

The boat is six point masses. With x the fraction of L forward of the transom and z the height above the keel:

Component i	Mass m_i (lb)	Position x_i	Height z_i (in)
hull	m_{hull} (2)	0.45	$\zeta_h D$
payload	$m_{\text{pay}} = 30$	ξ_p	$z_{\text{pay}} = 8$
battery	m_{bat} (3)	ξ_b	$z_{\text{bat}} = 4$
motor	m_{mot} (5)	0.10	$z_{\text{mot}} = 4$
propeller	m_{prop} (9)	0.00	$z_{\text{mot}} = 4$
fixed equipment	$m_{\text{fix}} = 8.8$	0.40	$z_{\text{fix}} = 8$

$$m = \sum_i m_i, \qquad \frac{\text{LCG}}{L} = \frac{\sum_i m_i x_i}{m} \ \text{(LCG in ft)}, \qquad \text{VCG} = \frac{\sum_i m_i z_i}{m} \ \text{[in]}, \qquad r_g = \kappa_g L. \tag{12}$$

The 30 lb payload and the fixed equipment are constants, so total mass moves with the hull, battery, motor and propeller diameter.

A.5 Hydrostatics (navaltoolbox on the hull mesh)

Freeboard is measured at rest, which is conservative because the boat rides higher when planing. The mesh from the previous subsection is handed to navaltoolbox's `HydrostaticsCalculator` (`hydrostatics` in the notebook). `from_displacement` finds the draft T_{stat} at which the immersed volume of the mesh weighs as much as the boat ($\rho_w V = m$), with the boat held level and the vertical center of gravity VCG from the mass model. From that equilibrium the library reports the waterplane, the wetted surface, the center of buoyancy and the transverse metacentric height

$$\text{GM} = \text{KB} + \text{BM} - \text{KG}, \qquad \text{BM} = I_T / V, \tag{13}$$

with I_T the transverse moment of inertia of the waterplane. The quantities used downstream are

$$\text{draft} = T_{\text{stat}}, \qquad \text{freeboard} = D - T_{\text{stat}} \ \text{[in]}, \qquad \text{GM [m]}.$$

If the solver finds no equilibrium (the hull cannot displace the boat's weight), the design is marked sunk: freeboard 0 and GM $= -1$, so the `freeboard` and `stability` margins both fail. Results are cached per hull geometry and loading, so the speed scan pays for one hydrostatics solve per design. The charts (not the margins) additionally use a free-trim solve, which lets the boat pitch until the center of buoyancy comes under the center of gravity and reports the at-rest trim.

A.6 Planing equilibrium (OpenPlaning)

For a given speed the boat is assumed to run at the trim and wetted length where lift balances weight and pitching moments balance. The notebook builds an OpenPlaning `PlaningBoat` from SI versions of the quantities above:

$$U = 0.44704\, v, \qquad W = 4.44822\, m, \qquad \text{beam} = 0.0254\, b, \qquad \text{lcg} = 0.3048\, \text{LCG}_{\text{ft}},$$
$$\text{vcg} = 0.0254\, \text{VCG}, \qquad r_g = 0.3048\, \kappa_g L, \qquad \rho = 16.01846\, \rho_w, \qquad v_T = 0.0254 \cdot 4,$$

with deadrise β, thrust angle $\varepsilon = 0$, thrust longitudinal offset 0 and kinematic viscosity ν. (The wave height H_w is passed as `H_sig`, but the notebook only calls the steady-trim and force routines, which do not use it; wave height therefore affects the model only through the freeboard allowance.)

Let $F_n = U/\sqrt{g\, b}$ be the beam speed coefficient (the notebook's `cv`) and $\lambda = (L_K + L_C)/(2b)$ the mean wetted length over beam, where L_K and L_C are the keel and chine wetted lengths that

follow from the trim τ and the waterline height. OpenPlaning's hydrodynamic force is Savitsky's planing-lift relation:

$$\begin{aligned} C_{L0} &= \tau^{1.1}\Big(0.012\,\lambda^{1/2} + 0.0055\,\frac{\lambda^{5/2}}{F_n^2}\Big), & C_{L\beta} &= C_{L0} - 0.0065\,\beta\,C_{L0}^{0.6}, \\ F_z &= C_{L\beta}\,\tfrac{1}{2}\rho U^2 b^2, & F_x &= F_z\tan\tau, \\ l_p &= \lambda b\Big(0.75 - \frac{1}{5.21\,(F_n/\lambda)^2 + 2.39}\Big), & M_{\mathrm{cg}} &= -\frac{F_z}{\cos\tau}\,(\mathrm{lcg} - l_p). \end{aligned} \tag{14}$$

Skin friction is added using the ITTC-1957 friction line plus a hull-roughness allowance (OpenPlaning default average roughness $150\,\mu$m): $R_f = \frac{1}{2}\rho\,(C_f + \Delta C_f)\,S_w U^2$ with wetted bottom area S_w, contributing $R_f\cos\tau$ to the horizontal force. Air drag, flaps and the roughness lift change are left at their zero defaults.

The solver (`get_steady_trim`, Newton iteration, trim limited to $[0.5°, 35°]$) finds the waterline height and trim τ for which the net vertical force and the net pitching moment about the centre of gravity are both zero, including the thrust's vertical component and its moment about the CG. The thrust the propeller must supply is then

$$T = \Big|\sum F_x\Big| = F_z\tan\tau + R_f\cos\tau \quad [\mathrm{N}], \tag{15}$$

the horizontal resultant of the water forces. The outputs used downstream are τ (`trim_deg`), T (`thrust_n`), L_K (`keel_m`), λ (`lam`) and $C_v = U/\sqrt{9.80665\,b}$ (`cv`). If OpenPlaning raises (no equilibrium) or T is not finite, the planing solve is marked failed (`ok = False`) and the design cannot be evaluated at that speed.

A.7 Race time

Race time uses speed over the ground, $v - c$, and charges the allowance t_a at cruise power:

$$T_{\mathrm{race}}(v) = \begin{cases} 3600\,\dfrac{D_c}{v-c} + t_a & \texttt{"worst"}\ \text{(default)}, \\ 3600\Big(\dfrac{D_c/2}{v-c} + \dfrac{D_c/2}{v+c}\Big) + t_a & \texttt{"outback"}, \end{cases} \qquad T_{\mathrm{race}} = \infty \text{ if } v \le c. \tag{16}$$

Here v is the speed through the water. With the defaults, $D_c = 2\,\mathrm{mi}$, $c = 1\,\mathrm{mph}$, $t_a = 20\,\mathrm{s}$.

A.8 Power chain and energy

$$P_{\mathrm{shaft}} = \frac{T\cdot 0.44704\,v}{\eta_p(T,v)}\ [\mathrm{W}], \qquad P_{\mathrm{bus}} = \frac{P_{\mathrm{el}}}{\eta_d} + P_h\ [\mathrm{W}], \qquad E_{\mathrm{used}} = \frac{P_{\mathrm{bus}}\,T_{\mathrm{race}}}{3600}\ [\mathrm{Wh}]. \tag{17}$$

The propeller efficiency η_p (11) depends on the thrust, the speed, the diameter and the pitch, so both propeller choices change the power. The motor's electrical power P_{el} comes from (7), and η_d is the speed controller's efficiency (Appendix B).

A.9 Battery under load

Solving $V = V_{\mathrm{oc}} - I\,R_{\mathrm{pack}}$ with $I = P_{\mathrm{bus}}/V$ gives $V^2 - V_{\mathrm{oc}}V + R_{\mathrm{pack}}P_{\mathrm{bus}} = 0$, whose larger root is

$$V_{\mathrm{load}} = \frac{V_{\mathrm{oc}} + \sqrt{V_{\mathrm{oc}}^2 - 4\,R_{\mathrm{pack}}\,P_{\mathrm{bus}}}}{2}, \qquad \text{defined only if } V_{\mathrm{oc}}^2 \ge 4\,R_{\mathrm{pack}}\,P_{\mathrm{bus}}, \tag{18}$$

otherwise the pack cannot deliver that power and V_{load} is undefined (`nan`). The remaining battery quantities are

$$I = \frac{P_{\text{bus}}}{V_{\text{load}}}, \qquad s_{\text{end}} = s_0 - \frac{E_{\text{used}}}{E_{\text{pack}}}, \qquad V_{\text{end}} = V_{\text{load}}\Big|_{V_{\text{oc}} \leftarrow S\cdot\text{OCV}(\max(s_{\text{end}},0))}. \tag{19}$$

That is, V_{end} is the root (18) with the open-circuit voltage replaced by its value at the end-of-race charge. Discharge is modelled at constant bus power and state of charge falls linearly with energy used. There is no thermal or rate-dependent capacity model.

A.10 Index: what each calculated value depends on

Table 3: Calculated values and the decision variables they depend on. "$*$" means all eleven searched variables.

Quantity	Equation	Depends on
h_c, A_{shell}, m_{hull}	(1), (2)	$L,\ b,\ \beta,\ D$ (and h_c only on b, β)
E_{pack}, m_{bat}, R_{pack}, I_{lim}	(3)	$S,\ n_p$
s_0, V_{oc}	(4)	S only
m_{mot}	(5)	μ only
$Q,\ I_m,\ V_m,\ P_{\text{el}},\ \eta_m$	(6), (7)	$*$ and v
m_{prop}	(9)	d_p only
rpm, v_{tip}	(8)	$p,\ d_p$ and v
m, LCG, VCG	(12)	$L,\ b,\ \beta,\ D,\ \xi_p,\ \xi_b,\ \mu,\ d_p,\ S,\ n_p$
static draft, freeboard, GM	mesh (App. A.5)	m and VCG (so all of $*$), $L,\ b,\ \beta,\ D$
$\tau,\ T,\ \lambda,\ L_K,\ C_v$	(14)–(15)	$*$ and v
T_{race}	(16)	v only
$\eta_p,\ P_{\text{shaft}},\ P_{\text{bus}},\ E_{\text{used}}$	(11), (17)	$*$ and v
$V_{\text{load}},\ I,\ s_{\text{end}},\ V_{\text{end}}$	(18), (19)	$*$ and v

B The constants

The `Fixed` dataclass holds everything that is not a decision. It is listed here once so the equations in the next sections can refer to symbols. Items marked † are placeholders, not measured or sourced values. Above `Fixed`, the notebook's Modes cell holds four explicit switches – `MOTOR_LIMITS`, `ENFORCE_VALIDITY`, `INCLUDE_SUSPECT_R` and `COURSE_MODE` – echoed by the problem card; this document assumes their defaults (`"sheet"`, `True`, `False`, `"worst"`).

Table 4: Constants of the model (default `Fixed` and `Cell`).
† marks placeholders.

Symbol	Code name	Unit	Value	Meaning
Rules				
m_{pay}	`payload_lb`	lb	30	Removable payload, fixed
h_{rule}	`freeboard_rule_in`	in	3.0	Minimum freeboard

Symbol	Code name	Unit	Value	Meaning
V_{lim}	`voltage_limit_v`	V	55.5	Maximum pack voltage at full charge
Environment				
ρ_w	`water_lb_ft3`	lb/ft^3	62.4	Fresh-water density
ν	`water_nu`	m^2/s	1.14×10^{-6}	Kinematic viscosity
H_w	`wave_height_in`	in	4.0†	Wave height (to be set from the venue)
f_w	`wave_allowance_factor`	–	0.5	Freeboard must cover $f_w H_w$ above the rule
c	`current_mph`	mph	1.0†	Opposing current
–	`course_mode`	–	`"worst"`	Whole course against the current; `"outback"` gives equal legs. Set by the `COURSE_MODE` mode switch
Mission				
D_c	`course_miles`	mi	2.0	Course length
t_a	`allowance_s`	s	20	Extra time, charged at cruise power
P_h	`hotel_w`	W	30	Electronics load
Design margins				
ϵ_u	`usable_energy_fraction`	–	0.70	Fraction of start-of-race energy that may be used
ρ_r	`rating_fraction`	–	0.80	Fraction of motor and cell ratings that may be used
$[\ell_-, \ell_+]$	`lcg_band`	–	[0.30, 0.45]	Allowed LCG/L from the transom
$V_{c,\text{min}}$	`min_cell_voltage_end`	V	3.0	Per-cell voltage floor at the finish
GM_{min}	`min_gmt_m`	m	0.05†	Required metacentric height at rest
Hull construction (synthetic)				
ρ_{lam}	`laminate_lb_ft2`	lb/ft^2	0.41†	Shell laminate mass per area
m_{str}	`hull_structure_lb`	lb	4.4†	Structure allowance
m_{fix}	`fixed_equipment_lb`	lb	8.8†	Aggregate fixed equipment
f_b	`bow_fraction`	–	0.25†	Forward fraction of L over which the chine beam tapers to the stem
Heights above the keel and inertia				
z_{pay}	`payload_z_in`	in	8†	Payload height
z_{bat}	`battery_z_in`	in	4†	Battery height
z_{mot}	`motor_z_in`	in	4†	Motor and propeller height
z_{fix}	`fixed_z_in`	in	8†	Fixed-equipment height

Symbol	Code name	Unit	Value	Meaning
ζ_h	`hull_z_frac_of_depth`	–	0.45†	Hull centre height $= \zeta_h D$
κ_g	`gyradius_frac`	–	0.25†	Radius of gyration $= \kappa_g L$
v_T	`thrust_height_in`	in	4.0†	Thrust line height above the keel
Propeller				
η_b	`prop_blade_eff`	–	0.70†	Blade, swirl and tip losses on top of the ideal disk efficiency (set so the default design lands near the old constant 0.65)
$\mathrm{pd}_{\mathrm{opt}}$	`prop_pd_opt`	–	1.2†	Pitch-to-diameter ratio where blade efficiency peaks
k_{pd}	`prop_pd_penalty`	–	0.15†	Efficiency lost per unit of $(p/d_p - \mathrm{pd}_{\mathrm{opt}})^2$
m_{ref}	`prop_mass_ref_lb`	lb	0.5†	Mass of a propeller of the reference diameter
d_{ref}	`prop_ref_diam_in`	in	6.0†	Reference diameter for the mass law
$v_{\mathrm{tip,max}}$	`max_tip_speed_fps`	ft/s	150†	Blade tip-speed limit (cavitation and noise)
Motor and drive				
η_d	`drive_eff`	–	0.97†	Speed controller efficiency, battery bus to motor terminals
σ_s	`slip_factor`	–	0.85	Slip factor in the rpm equation
Cell (generic 21700, `Cell`)				
q_c	`ah`	Ah	5.0†	Cell capacity
V_{nom}	`v_nominal`	V	3.6†	Nominal cell voltage
V_{full}	`v_full`	V	4.2†	Full-charge cell voltage
r_c	`resistance_ohm`	Ω	0.015†	Cell internal resistance
m_c	`mass_lb`	lb	0.154†	Cell mass
$I_{c,\mathrm{max}}$	`max_current_a`	A	15†	Continuous current per cell
κ_o	`pack_overhead`	–	0.15†	Case, BMS and wiring as a fraction of cell mass

The open-circuit-voltage curve is a generic NMC table†, interpolated linearly:

State of charge s	0	0.1	0.2	0.3	0.5	0.7	0.8	0.9	1.0
Cell OCV(s) (V)	3.00	3.45	3.55	3.65	3.75	3.92	4.00	4.10	4.20

Component positions that are hard-coded inside `evaluate()` (fractions of L forward of the transom):

hull 0.45, motor 0.10, propeller 0.00, fixed equipment 0.40. Only the payload and battery positions are decisions.

Motor data. The motor itself is not in `Fixed`. It is one row of `sss_motors.csv`, which was typed from the supplier's spec sheet (27 motor models, 263 windings). A winding is `r_suspect` when $I_{m,\max} R_m / V_{m,\max}$ is outside $[0.01,\ 0.30]$, and those 33 windings are left out of the catalog unless `INCLUDE_SUSPECT_R = True`, leaving 230. The baseline winding, 56154-V2 5Y, has $K_v = 400$ rpm/V, $I_{m,\max} = 320$ A, $V_{m,\max} = 62$ V, $R_m = 0.0165\,\Omega$, $I_0 = 3.5$ A, $P_{\text{cont}} = 14000$ W and a mass of 3.92 lb.

Unit conversions: $1\,\text{lbf} = 4.44822162\,\text{N}$, $1\,\text{in} = 0.0254\,\text{m}$, $1\,\text{ft} = 0.3048\,\text{m}$, $1\,\text{mph} = 0.44704\,\text{m/s}$, $1\,\text{lb/ft}^3 = 16.01846\,\text{kg/m}^3$.

C The twenty-four margins

Each check is a margin $g_k(x, v)$, positive when it passes, with a scale σ_k. A design at speed v is *feasible* if

$$g_k(x, v) \geq 0 \quad \text{for all } k = 1, \ldots, 24, \tag{20}$$

and its *total violation* (used only to steer the search toward feasibility) is

$$\text{viol}(x, v) = \sum_k \max\bigl(0,\ -g_k/\sigma_k\bigr). \tag{21}$$

The margins fall in four groups (`rule`, `limit`, `geometry`, `validity`). The `validity` group is switchable through `ENFORCE_VALIDITY`, which is `True` by default and is assumed throughout this document.

Table 5: All 24 margins. "Depends on v" says whether the margin changes with speed.

Code name	Group	Margin $g_k \geq 0$	Scale σ_k	Depends on v
Rules				
`freeboard`	rule	$D - T_{\text{stat}} - h_{\text{rule}} - f_w H_w$ (T_{stat} in inches, from the mesh)	3.0	no
`ground_speed`	rule	$v - c$	5.0	only v
`voltage_rule`	rule	$V_{\text{lim}} - V_{\text{oc}}$	V_{lim}	no (only S)
Hull geometry				
`depth_over_chine`	geometry	$D - 12\,h_c$	12.0	no
`prop_fits`	geometry	$D - d_p$	D	no
Propulsion, battery and balance limits				
`motor_max_voltage`	limit	$V_{\text{cap}} - S\,V_{\text{full}}$	V_{cap}	no (only $S,\ \mu$)

continued on the next page

Code name	Group	Margin $g_k \geq 0$	Scale σ_k	Depends on v
`energy`	limit	$\epsilon_u\, E_{\text{pack}}\, s_0 - E_{\text{used}}$	$\max(\epsilon_u E_{\text{pack}} s_0,\ 1)$	yes
`shaft_power`	limit	$\rho_r\, P_{\text{cont}} - P_{\text{shaft}}$	P_{cont}	yes
`battery_current`	limit	$\rho_r\, I_{\text{lim}} - I$	I_{lim}	yes
`motor_current`	limit	$\rho_r\, I_{\text{cap}} - I_m$	I_{cap}	yes
`motor_voltage`	limit	$V_{\text{load}} - V_m$ (-1 if V_{load} undefined)	V_{cap}	yes
`pack_can_deliver`	limit	$+1$ if $V_{\text{oc}}^2 \geq 4R_{\text{pack}}P_{\text{bus}}$, else -1	1	yes
`end_cell_voltage`	limit	$V_{\text{end}}/S - V_{c,\text{min}}$ (-1 if V_{end} undefined)	1	yes
`tip_speed`	limit	$v_{\text{tip,max}} - v_{\text{tip}}$	$v_{\text{tip,max}}$	yes
`lcg_min`	limit	$\text{LCG}/L - \ell_-$	0.1	no
`lcg_max`	limit	$\ell_+ - \text{LCG}/L$	0.1	no
`stability`	limit	$\text{GM} - \text{GM}_{\text{min}}$	0.1	no
Planing-method validity (Savitsky's applicability limits)				
`solver_ok`	validity	$+1$ if the planing solve succeeded, else -1	1	yes
`trim_min`	validity	$\tau - 2^\circ$	2	yes
`trim_max`	validity	$15^\circ - \tau$	15	yes
`wetted_length_ratio`	validity	$4 - \lambda$	4	yes
`cv_min`	validity	$C_v - 0.6$	0.6	yes
`cv_max`	validity	$13 - C_v$	13	yes
`keel_within_hull`	validity	$0.3048\, L - L_K$	$0.3048\, L$	yes

Here are the same margins for the default design at its top speed.

Table 6: Margins of the baseline at $v^\star = 21.25$ mph. The last column is g_k/σ_k. `voltage_rule` is 0 by construction (see "What to doubt"), so the binding constraint is the smallest ratio among the rest: `trim_min`.

Constraint	Group	Margin g_k	Scale σ_k	g_k/σ_k
`freeboard`	rule	4.567	3	1.522
`stability`	limit	0.1582	0.1	1.582
`depth_over_chine`	geometry	10.59	12	0.882
`prop_fits`	geometry	6	12	0.500
`ground_speed`	rule	20.25	5	4.050
`voltage_rule`	rule	0	55.5	0.000
`motor_max_voltage`	limit	3.2	62	0.052
`energy`	limit	544.5	799.5	0.681
`shaft_power`	limit	9728	14000	0.695
`battery_current`	limit	26.66	90	0.296
`motor_current`	limit	29.47	320	0.092
`motor_voltage`	limit	43.58	62	0.703
`pack_can_deliver`	limit	1	1	1.000
`end_cell_voltage`	limit	0.7059	1	0.706
`tip_speed`	limit	80.88	150	0.539
`lcg_min`	limit	0.04649	0.1	0.465
`lcg_max`	limit	0.1035	0.1	1.035
`solver_ok`	validity	1	1	1.000
`trim_min`	validity	0.003158	2	**0.002**
`trim_max`	validity	13	15	0.866
`wetted_length_ratio`	validity	0.6399	4	0.160
`cv_min`	validity	4.158	0.6	6.931
`cv_max`	validity	8.242	13	0.634
`keel_within_hull`	validity	0.8958	2.591	0.346